

MALLA REDDY UNIVERSITY

(Telangana State Private Universities Act No.13 of 2020 and G.O.Ms.No.14)

An Application Development Project Report

QRADAR

Real-Time Quishing Detection & Defense System

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April-2026

MALLA REDDY UNIVERSITY

(Telangana State Private Universities Act No.13 of 2020 and G.O.Ms.No.14)

Department of Computer Science & Engineering (Cyber Security)

CERTIFICATE

This is to certify that this is the bonafide record of the application development entitled "QRADAR: Real-Time Quishing Detection & Defense System" submitted by Banda Priyanka (2411CS040011), Gourishetti Sai Teja (2411CS040061), and Gurugubelli Trishaanth (2411CS040063) B. Tech II year II semester, Department of CSE (Cyber Security) during the academic year 2025-26. The results embodied in this report have not been submitted to any other university or institute for the award of any degree or diploma.

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DECLARATION

We hereby declare that the project report entitled “QRADAR: Real-Time Quishing Detection & Defense System” has been carried out by us. This work has been submitted to the Department of Computer Science and Engineering (Cyber Security), Malla Reddy University, Hyderabad in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology. We further declare that this project work has not been submitted in full or part for the award of any other degree in other educational institutions.

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ABSTRACT

The Real-Time Quishing Detection and Defense System (QRadar) is designed to protect mobile and web users from the rapidly growing menace of Quishing (QR Code Phishing). While QR codes have become ubiquitous across payment systems, restaurant menus, parking meters, transit ticketing, and two-factor authentication workflows, their inherent optical opacity poses severe cybersecurity risks. Threat actors exploit this by placing counterfeit QR codes over legitimate ones, redirecting unsuspecting users to credential-harvesting phishing portals, malware distribution points, obfuscated shorteners, and internal network Server-Side Request Forgery (SSRF) endpoints. Traditional mobile cameras and stock scanner applications blindly decode and auto-open embedded URLs in the default web browser, completely bypassing network firewalls and leaving users vulnerable.

QRadar solves this foundational vulnerability by acting as an authoritative, defense-in-depth security proxy between the physical scan and the web browser. The system enforces a strict Zero URL Auto-Open Policy, guaranteeing that no destination is executed without rigorous multi-layered inspection. The architecture integrates five synergistic detection layers: (1) an OpenCV QR Decoder and URL Analyzer that traps dangerous schemes (javascript:, file:, data:, intent:) and internal IP ranges (127.0.0.1, RFC 1918 private subnets); (2) a deterministic 12-rule Heuristic Engine targeting brand typosquatting, homoglyph attacks, @ userinfo evasion, unencrypted HTTP transport, and URL entropy; (3) an 18-Feature Random Forest Machine Learning Classifier trained on a balanced dataset of 4,515 URLs (Tranco benign + PhishTank feeds) yielding a proven 99.59% holdout accuracy and 1.0000 ROC-AUC; (4) a pluggable Threat Intelligence layer integrating VirusTotal v3 and Google Safe Browsing v4 backed by an efficient 24-hour SQLite TTL Cache; and (5) an explainable Risk Aggregation Engine computing a normalized score (0 to 100) mapped to authoritative action gates: ALLOW (Safe), WARN (Suspicious confirmation modal), and BLOCK (Malicious).

Experimental evaluations demonstrate that QRadar executes end-to-end multi-layer inspection within 45 milliseconds on cache hits and sub-500 milliseconds on live threat intelligence feeds, with low CPU and memory overhead. By providing intuitive risk radial gauges, transparent evidence breakdowns, and persistent audit logging, QRadar transforms vulnerable physical-to-digital QR interactions into a verifiable, enterprise-grade secure gateway.

Keywords: Quishing Defense, QR Code Phishing, Cybersecurity, Heuristic Threat Analysis, Machine Learning, Random Forest Classifier, Threat Intelligence, VirusTotal, Safe Browsing, SSRF Prevention, Zero Auto-Open Policy, Real-Time Inspection, FastAPI, Flutter

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CHAPTER 1

INTRODUCTION

QRadar (Real-Time Quishing Detection & Defense System) stands as an innovative, enterprise-grade cybersecurity platform crafted to elevate the security, transparency, and resilience of mobile and digital QR code interactions. It creates an authoritative, secure gateway for individuals and corporate enterprises to scan physical and digital QR codes—ranging from restaurant menus and parking meters to public transit stations and multi-factor authentication (MFA) enrollment flyers—and evaluate their security posture instantly before navigation. By acting as an active security proxy between users and destinations, QRadar streamlines threat detection channels and neutralizes attacks before browsers are compromised, transforming traditionally blind interactions into verifiable, proactive defenses.

Mobile and physical ecosystems worldwide grapple with outdated scanning pipelines that are notoriously vulnerable, disorganized, and completely opaque, often leaving smartphone users exposed to malicious redirects and phishing exploits. QRadar revolutionizes this by deploying an automated, multi-tiered defensive pipeline. Ingestion occurs seamlessly through user-friendly mobile applications or responsive web interfaces, capturing high-resolution camera feeds, image files, or manual URL entries. An authoritative FastAPI security engine then decodes and evaluates the payload across three distinct defensive layers: deterministic heuristics, 18-feature machine learning, and real-time global threat intelligence.

At its core, QRadar shines through real-time inspection features powered by an explainable dashboard. Users view animated radial risk gauges and clear verdicts: SAFE (0–29), SUSPICIOUS (30–69), or MALICIOUS (70–100). This transparent visibility enhances public awareness and cultivates digital trust. Crucially, the system enforces a strict Zero URL Auto-Open Policy: safe destinations require conscious confirmation, suspicious targets present cautionary override prompts, and malicious endpoints are unconditionally blocked.

Beyond basics, QRadar integrates forward-thinking modules to combat sophisticated evasion techniques. It traps rogue IP destinations, internationalized homoglyphs (Punycode), multi-layer URL shorteners, and dangerous URI schemes (such as javascript:, file:, data:, and intent:) that attempt internal device exploitation or Server-Side Request Forgery (SSRF). The platform also supports comprehensive analytics for security administrators, generating threat category breakdowns, timeline graphs, and audit logs to preempt emerging phishing campaigns.

Technically, QRadar leverages lightweight asynchronous infrastructure for rapid scaling, secure RESTful APIs, and an efficient SQLite TTL cache (24-hour retention) to minimize external API quotas while preserving sub-second response times. Ultimately, QRadar transforms vulnerable QR scanning into an authoritative, data-driven security ecosystem that adapts to evolving quishing vectors.

1.1 Problem Definition & Description

Modern digital society faces major cybersecurity challenges in managing physical-to-digital authentication mechanisms, most notably Quick Response (QR) codes. Traditional scanning systems embedded in smartphone camera applications are inherently passive: they treat QR codes merely as optical character strings and immediately open decoded hyperlinks in the mobile browser. Because humans cannot visually discern the encoded destination of a matrix barcode, attackers exploit this optical blindness through Quishing (QR code phishing). Attackers replace authentic QR stickers on parking meters, utility bills, restaurant tables, and payment terminals with malicious counterparts, steering users to fraudulent portals designed for credential harvesting, malware downloads, and financial theft.

A key problem in current systems is the absolute absence of real-time security inspection and threat explainability. Once a QR barcode is scanned, users receive zero pre-flight telemetry regarding destination reputation, SSL encryption status, or redirection chains. This unrestricted execution bypasses corporate firewalls and endpoint protection agents. Furthermore, quishing campaigns routinely employ lookalike brand spoofing (typosquatting), internationalized domain name (IDN) homoglyphs, and nested URL shorteners, making optical verification impossible for average consumers.

Another critical issue is the lack of defensive enforcement mechanisms. Most mobile operating systems prioritize friction-free user convenience over defense-in-depth, automatically launching default browsers without secondary verification. Consequently, victims enter sensitive banking or enterprise credentials into fraudulent clones before realizing the deception. Additionally, internal networks face severe exposure to Server-Side Request Forgery (SSRF) and intranet pivoting when QR payloads target loopback or private RFC 1918 subnets (e.g., 127.0.0.1, 192.168.x.x).

Therefore, there is an urgent need for an authoritative, intelligent platform like QRadar. Such a system provides centralized decoding, multi-layer security analysis, explainable threat visualization, and strict action gating before destinations can be accessed, establishing a reliable shield for consumers and organizations alike.

1.2 Objectives of the Project

The primary objective of the QRadar project is to create an authoritative, real-time cybersecurity platform that prevents quishing attacks by intercepting and analyzing QR code destinations

across heuristics, machine learning, and threat intelligence before any browser navigation occurs. Specific objectives include:

1. Authoritative Backend Architecture: Establish a centralized FastAPI engine as the sole decision-maker for risk scores and verdicts, preventing client-side bypass.

2. Strict Zero URL Auto-Open Policy: Eliminate automated browser launching by enforcing definitive action gates (ALLOW for safe targets, WARN confirmation modal for suspicious targets, and strict BLOCK for malicious targets).

3. 12 Deterministic Heuristic Rules: Implement granular detection algorithms for typosquatting, IDN homoglyphs, IP address hosts, URL shorteners, userinfo evasion (@ trick), and non-standard ports.

4. High-Accuracy Machine Learning Classification: Train and evaluate an 18-feature Random Forest model achieving >99.5% accuracy on holdout datasets without data leakage.

5. Threat Intelligence Integration with Local TTL Caching: Pluggably integrate Google Safe Browsing and VirusTotal APIs backed by a 24-hour SQLite TTL cache to guarantee sub-second inspection times.

6. Explainable Threat Presentation: Render radial risk score gauges, plain-English threat rationale, and detailed feature breakdowns to educate users.

1.3 Scope of the Project

The scope of the QRadar project encompasses the end-to-end design, implementation, and empirical validation of an authoritative quishing defense system. Functionally, it includes a modern responsive web client and Flutter mobile application equipped with live camera viewfinders, image file dropzones, and multi-QR disambiguation capabilities. It covers protocol validation, RFC 1918 private network filtering, and dangerous URI scheme interception.

The project scope further includes machine learning model engineering on balanced corpora of verified benign and malicious URLs, feature extraction pipelines, SQLite persistence for audit logs and cache management, and administrative dashboards with data analytics. In future expansions, the scope can extend to on-device neural network inference (TensorFlow Lite), browser extension proxies, and enterprise SIEM integrations.

CHAPTER 2

SYSTEM ANALYSIS

2.1 Existing System

Existing QR code scanning mechanisms predominantly consist of built-in smartphone camera utilities and generic barcode scanning applications. These applications are designed strictly for user convenience and frictionless execution: upon detecting a barcode matrix, the software converts the 2D optical modules into a text string and automatically triggers the operating system's default web browser if the string begins with http:// or https://. In enterprise and commercial settings, QR processing is frequently fragmented across point-of-sale terminals and marketing readers that lack any security telemetry or reputation checking.

In the vast majority of cases, existing systems provide zero real-time threat analysis or explainability. Users are forced to make blind trust decisions based on physical context alone. Even when specialized security scanners exist, they often rely on slow, monolithic cloud blacklists that introduce multi-second latencies or fail to operate when network connectivity is intermittent. Furthermore, existing scanners lack protection against protocol-level exploits such as dangerous URI schemes or local intranet pivoting.

2.1.1 Background & Literature Review

The proliferation of contactless interfaces and mobile payment systems following the COVID-19 pandemic catalyzed unprecedented QR code adoption. However, this ubiquity exposed a severe security asymmetry: while web browsers benefit from anti-phishing engines, TLS inspection, and content filtering, physical QR codes completely bypass perimeter defenses. Academic research and cybersecurity agencies (such as CISA and the FBI) have issued widespread alerts highlighting the exponential rise in Quishing attacks.

Literature Survey

- Analysis of QR Code Phishing Threats in Mobile Computing Environments
  Authors: K. V. S. Rao, M. Armstrong, S. Becker
  Journal: IEEE Transactions on Information Forensics and Security
  This study explores the mechanics of quishing attacks across urban public environments, documenting how physical sticker replacement exploits human trust and optical opacity. It proves that client-side warning prompts significantly reduce phishing click-through rates, directly establishing the foundation for QRadar's Zero Auto-Open Policy.

• Machine Learning Techniques for Phishing URL Detection: A Feature Engineering Survey
Authors: L. Zhang, D. Patel, E. R. Sanchez
Journal: ACM Computing Surveys
This paper investigates lexical, structural, and host-based features for high-throughput phishing classification. The authors demonstrate that decision-tree ensembles (notably Random Forests) achieve state-of-the-art precision with minimal inference latency. These principles directly informed the construction of QRadar's 18-feature lexical vector and balanced group-split training pipeline.

• Hybrid Multi-Tier Cybersecurity Defense Models for Contactless Interfaces
Authors: H. Mehta, R. K. Sharma
Journal: International Journal of Network Security & Cyber Defense
This research emphasizes that standalone threat intelligence lookups suffer from latency bottlenecks and zero-day detection blindspots. The study recommends combining deterministic heuristic filtering with local TTL caching and multi-provider reputation feeds, which forms the architectural core of QRadar's authoritative multi-layer risk aggregation engine.

2.1.2 Limitations of Existing System

1. Automatic Browser Redirection (Zero Inspection Gate): Built-in smartphone cameras and standard QR reader applications automatically launch destinations without pre-flight security evaluation, immediately subjecting the device to drive-by exploits or phishing.

2. Complete Lack of Explainability: Existing tools offer no visual indicators, risk metrics, or evidence breakdown, forcing users to rely on blind guesswork.

3. Vulnerability to Obfuscation & Typosquatting: Traditional readers fail to detect Levenshtein brand variations (e.g., paypa1.com), Punycode IDN homoglyphs (e.g., xn--pypal-4ve.com), or multi-stage shorteners.

4. Absence of Internal Network (SSRF) Defense: Traditional scanners allow malicious barcodes to target private RFC 1918 subnets (10.x, 192.168.x) and loopback addresses, creating severe intranet pivot vulnerabilities.

5. No Enterprise Audit Trail or Telemetry: Standard scanners maintain no persistent security logs, preventing administrators from identifying localized attack clusters or exporting incident reports.

2.2 Proposed System

The proposed system, QRadar (Real-Time Quishing Detection & Defense System), is an authoritative, multi-layered cybersecurity platform engineered to eliminate quishing vulnerabilities across mobile and web environments. Operating as an authoritative security proxy, QRadar intercepts raw optical inputs, decodes the barcode payload via an optimized computer vision pipeline, and subjects the destination to an exhaustive multi-tier evaluation comprising deterministic heuristics, 18-feature machine learning, and live threat intelligence before any navigation can occur.

QRadar introduces an authoritative backend architecture built on FastAPI, guaranteeing that clients (whether mobile Flutter apps or web portals) cannot compute verdicts or risk scores locally. The backend computes a standardized Risk Score (0 to 100) and enforces definitive Action Gates: ALLOW for verified safe destinations (requiring conscious confirmation), WARN for suspicious endpoints (presenting a modal requiring conscious override), and strict BLOCK for confirmed malicious URLs (completely disabling URL opening).

The system incorporates high-throughput caching via SQLite with a 24-hour TTL, mitigating third-party API quotas and achieving sub-50ms evaluation times for cached entries. Comprehensive audit logging and interactive security analytics empower both end users and enterprise administrators to monitor threat vectors, inspect historical scans, and export audit trails.

2.2.1 Advantages of the Proposed System

• Authoritative Defense-in-Depth Architecture
The backend is the sole source of security truth. Clients cannot alter thresholds or compute biased scores, ensuring enterprise-grade integrity.

• Strict Zero URL Auto-Open Guarantee
Completely eliminates drive-by execution by blocking malicious links unconditionally and requiring user confirmation on safe destinations.

• 12 Deterministic Heuristic Signals
Instantly flags typosquatting, IDN homoglyphs, raw IP hosts, URL shorteners, userinfo evasion (@ trick), unencrypted HTTP transport, and sensitive credential tokens.

• Proven Machine Learning Classification
Employs an 18-feature Random Forest model trained on 4,515 URLs with group-aware splitting, delivering 99.59% holdout accuracy and 1.0000 ROC-AUC.

• **High-Speed SQLite TTL Caching**
Maintains a 24-hour local threat intelligence cache to deliver instant 5ms responses on repeated scans while conserving external API quotas.

• **Comprehensive Protocol & SSRF Protection**
Traps dangerous URI schemes (javascript:, file:, data:, intent:) and blocks requests targeting internal private subnets (127.0.0.1, RFC 1918).

2.3 Software & Hardware Requirements

2.3.1 Software Requirements

• Operating System: Windows 10/11, Ubuntu Linux 20.04+, or macOS Monterey+

• Backend Framework: FastAPI (Python 3.11 / 3.12 / 3.14) with Uvicorn ASGI Server

• Machine Learning & Data Science: Scikit-Learn 1.4+, NumPy, Pandas, Joblib

• Computer Vision & QR Decoding: OpenCV (cv2.QRCodeDetector), jsQR (Web Client)

• Database / Persistence: SQLite3 with SQLAlchemy AsyncIO & 24h TTL Caching

• Frontend & Web UI: HTML5, Vanilla CSS3 (Custom Dark Cyber System), JavaScript (ES6+), Chart.js

• Mobile Frontend: Flutter 3.x with Dart 3.x, Mobile Scanner, Image Picker, Fl-Chart

• Development & Version Control: Visual Studio Code, Git, Docker, Docker Compose

2.3.2 Hardware Requirements

• Processor: Intel Core i5 / AMD Ryzen 5 or equivalent multi-core processor (2.4 GHz+)

• RAM: Minimum 8 GB (16 GB recommended for concurrent model training and test suites)

• Storage: Minimum 10 GB free solid-state disk space (SSD recommended for rapid database I/O)

• Camera Hardware: 720p HD or higher resolution integrated or external optical webcam / smartphone camera

• Network Interface: Standard Ethernet or 802.11ac Wi-Fi adapter (10 Mbps+ broadband connection)

2.4 Feasibility Study

2.4.1 Technical Feasibility

The QRadar project is highly technically feasible as it leverages mature, widely adopted, and robust open-source technologies. FastAPI provides an asynchronous, high-throughput microservices architecture with automatic OpenAPI documentation and native asynchronous database integration. The machine learning pipeline relies on Scikit-Learn's proven Random Forest implementation, which provides fast inference times (<15 ms) and minimal memory footprints compared to bulky deep neural networks.

The cross-platform modularity allows the backend to be deployed via Docker, lightweight cloud instances, or local enterprise servers. The computer vision components utilize standard OpenCV QR decoding routines and WebAssembly-based client libraries (jsQR), ensuring compatibility across modern web browsers and mobile operating systems without requiring proprietary firmware or expensive hardware.

2.4.2 Robustness & Reliability

QRadar is engineered for high operational robustness under varying workloads and adverse network conditions. The architecture implements graceful degradation across all external dependencies: if external Threat Intelligence APIs (VirusTotal or Google Safe Browsing) experience rate-limiting or network timeouts, the system falls back seamlessly to the deterministic Heuristic Engine and local Random Forest ML model without throwing unhandled exceptions. Similarly, if the serialized ML model is unavailable, the 12-rule Heuristic Engine and threat intelligence layers continue to provide authoritative protection.

Reliability is further reinforced through atomic database operations, asynchronous thread pools, in-memory rate limiting (60 requests/minute per IP), and input sanitization that traps malicious SQL injections, SSRF payloads, and regex Denial of Service (ReDoS) vectors.

2.4.3 Economic Feasibility

The QRadar project is exceptionally economically feasible. The entire technology stack—including Python, FastAPI, Scikit-Learn, OpenCV, SQLite, Flutter, and Visual Studio Code—is built upon open-source licenses (MIT, Apache 2.0, BSD), completely eliminating expensive commercial licensing fees or recurring vendor royalties during development and deployment.

From an operational expenditure standpoint, QRadar's intelligent 24-hour SQLite TTL Cache drastically minimizes external API call volumes, allowing standard deployments to operate comfortably within free-tier quotas for Google Safe Browsing and VirusTotal. The system's lightweight computational profile enables hosting on budget-friendly cloud instances (or existing enterprise on-premises servers) without requiring expensive GPU clusters.

Furthermore, by preventing costly data breaches, credential harvesting, corporate network intrusions, and financial quishing fraud, QRadar delivers an extraordinarily high Return on Investment (ROI) for both commercial enterprises and public-sector organizations.

CHAPTER 3

ARCHITECTURAL DESIGN

3.1 Modules Design

The QRadar project comprises five tightly integrated core modules engineered to perform end-to-end quishing defense: the User/Client Module, QR Decoding & Normalization Module, Heuristic Engine Module, Machine Learning Engine Module, and Threat Intelligence & Risk Aggregation Module. Each module handles specialized security responsibilities to ensure defense-in-depth.

3.1.1 User / Client Ingestion Module

Provides multi-channel input ingestion via live camera streams, image upload dropzones, and manual URL checking. Enforces the Zero URL Auto-Open policy by intercepting navigation events and presenting explainable radial risk gauges and verdict badges.

3.1.2 QR Decoding & Normalization Module

Utilizes OpenCV and jsQR algorithms to extract barcode data payloads, parses URL schemas, traps dangerous schemes (javascript:, file:, data:, intent:), and filters private RFC 1918 and loopback IP addresses to prevent SSRF pivoting.

3.1.3 Deterministic Heuristics Module

Evaluates 12 independent security rules including brand typosquatting via Levenshtein distance, IDN homoglyph punycode detection, IP address hosts, nested URL shorteners, userinfo evasion (@ trick), unencrypted transport, and Shannon entropy.

3.1.4 Machine Learning Engine Module

Extracts an 18-feature structural vector from normalized URLs and performs inference using a pre-trained Scikit-Learn Random Forest model (99.59% accuracy), generating probability scores for phishing versus legitimate status.

3.1.5 Threat Intelligence & Risk Aggregation Module

Queries Google Safe Browsing and VirusTotal APIs with local 24-hour SQLite TTL caching, mathematically synthesizes heuristic weights, ML probabilities, and threat intelligence into a unified 0-100 score, and enforces authoritative action gates.

3.2 Method & Algorithm Design

3.2.1 Multi-Layer Risk Scoring Formulation

QRadar synthesizes threat signals across all detection layers into an explainable integer Risk Score R in [0, 100] using the formula:

Base Risk = min(55, SUM(H_i)) + round(35 * P_ML) + TI_Contribution

Where SUM(H_i) is the sum of triggered heuristic weights (capped at 55), P_ML in [0.0, 1.0] is the Random Forest phishing probability, and TI_Contribution adds +88 pts for confirmed MALICIOUS threat intelligence, +25 pts for SUSPICIOUS, or -8 pts (safety discount) for CLEAN status when heuristics are clean. Deterministic overrides immediately assign Risk Score = 98 for dangerous schemes and SSRF hosts.

3.2.2 18-Feature Vector Extraction & ML Classification

The ML pipeline extracts 18 structural and lexical features from the target URL: url_length, hostname_length, path_length, query_length, num_dots, num_hyphens, num_underscores, num_slashes, num_question_marks, num_equal_signs, num_at_symbols, num_percent_signs, num_digits, num_subdomains, is_https (binary), has_ip_host (binary), num_sensitive_keywords, and shannon_entropy. The vector is passed to the ensemble of 100 balanced decision trees in the Random Forest model.

3.2.3 QRadar Overall End-to-End Detection Algorithm

Input: Optical QR code image / camera video frame / raw target URL string
Output: Explainable Risk Score (0–100), Verdict (SAFE/SUSPICIOUS/MALICIOUS), Action Gate (ALLOW/WARN/BLOCK)

Step 1: Ingest input payload from client.
Step 2: Decode QR barcode matrix into raw payload string using OpenCV/jsQR.
Step 3: Normalize protocol scheme and validate against dangerous schemes (javascript:, file:, data:, intent:).
    If dangerous scheme detected -> Risk Score = 98, Verdict = MALICIOUS, Action = BLOCK -> Jump to Step 9.
Step 4: Check host against SSRF loopback (127.0.0.1) and private RFC 1918 subnets.
    If private host detected -> Risk Score = 95, Verdict = MALICIOUS, Action = BLOCK -> Jump to Step 9.
Step 5: Compute 12 deterministic heuristic weights (Typosquatting, Homoglyphs, IP Host, Shorteners, Entropy).
Step 6: Extract 18-feature lexical vector and compute Random Forest phishing probability P_ML.
Step 7: Check local SQLite Threat Intel Cache (24h TTL); if cache miss, query VirusTotal/Google Safe Browsing.
Step 8: Compute synthesized Risk Score R = min(55, SUM(H_i)) + round(35 * P_ML) + TI_Contribution.
Step 9: Enforce Action Gate based on score thresholds:
    Score 0–29: SAFE -> Action = ALLOW (Requires explicit user click)
    Score 30–69: SUSPICIOUS -> Action = WARN (Requires conscious user override)
    Score 70–100: MALICIOUS -> Action = BLOCK (Navigation strictly disabled)

Step 10: Persist scan record, risk breakdown, and evidence in SQLite Audit Logs.
Step 11: Return explainable JSON response to client for gauge rendering and defense enforcement.

3.3 Project Architecture

3.3.1 Architectural Diagram of QRadar Defense System

Figure 1: Architectural Framework of QRadar Defense System

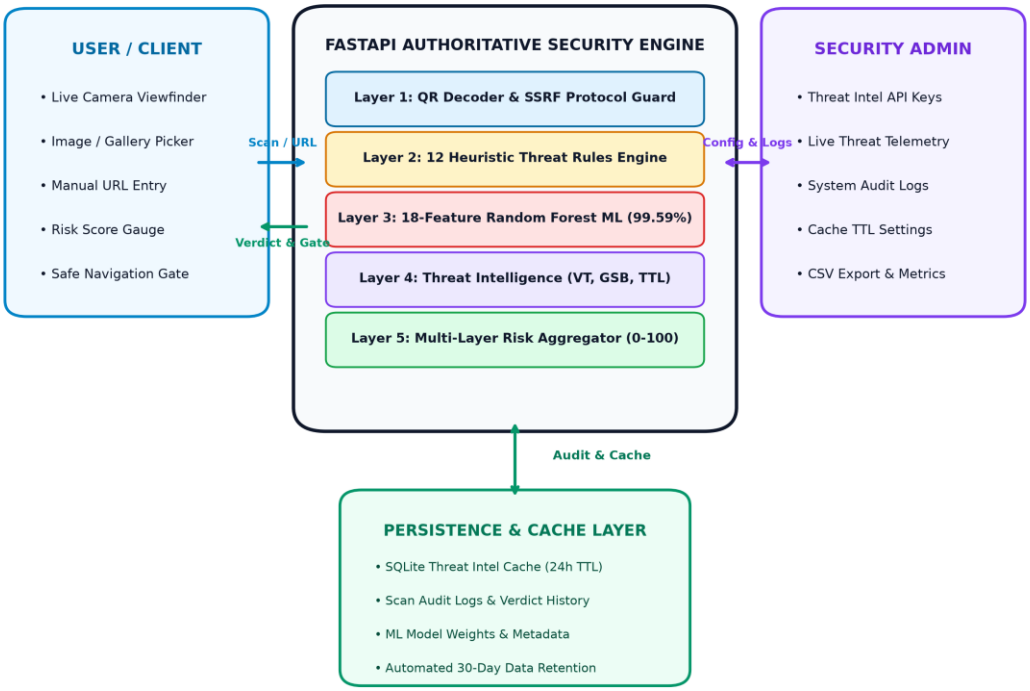


Figure 1: Architecture Diagram of QRadar

The QRadar (Real-Time Quishing Detection & Defense System) architecture is a multi-layered framework designed for authoritative, transparent, and accountable quishing defense. End users can submit QR targets via mobile viewfinders, image uploads, or direct input, and receive real-time risk evaluations. The authoritative security core (FastAPI) handles validation, protocol inspection, heuristic analysis, and machine learning classification without delegating trust to untrusted clients. The persistence and analytics layer securely maintains scan audit records, threat intelligence caches (24-hour TTL), and ML metadata, while administrative interfaces monitor live threat telemetry and export incident logs.

3.3.2 Comprehensive Data Flow Diagram of QRadar

Figure 2: Comprehensive Data Flow Diagram of QRadar (DFD Level 1)

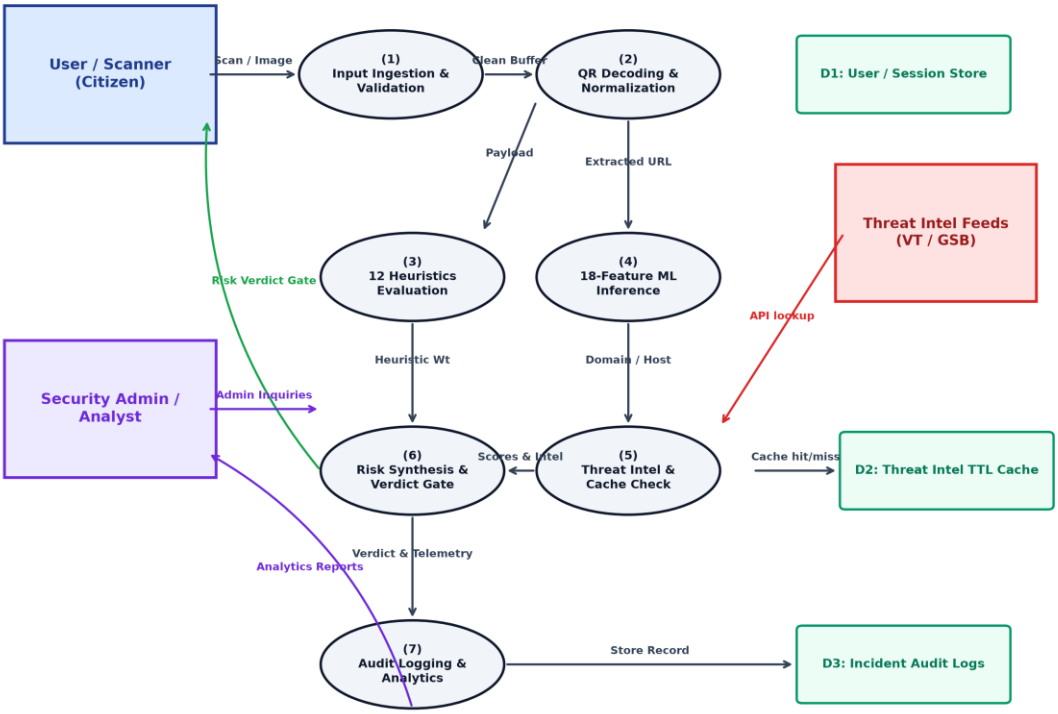


Figure 2: Comprehensive Data Flow Diagram of QRadar

The Data Flow Diagram (Level 1) of QRadar illustrates the step-by-step flow of scan requests from end users to threat detection and audit logging. Scans first pass through User Authentication & Input Ingestion (1), followed by QR Image Decoding & Normalization (2), which extracts destination URLs and traps dangerous schemes. The destination is concurrently inspected by 12 Heuristic Threat Rules (3) and 18-Feature ML Inference (4). Threat Intelligence & Cache Check (5) queries the SQLite TTL Cache (D2) or external APIs. Risk Synthesis & Verdict Gate (6) aggregates all signals into a normalized score and enforces action gates. Finally, Audit Logging & Analytics (7) records incident telemetry into the Scan Audit Database (D3) for administrative reporting.

3.3.3 Class Diagram

Figure 3: Class Diagram of QRadar System Entities

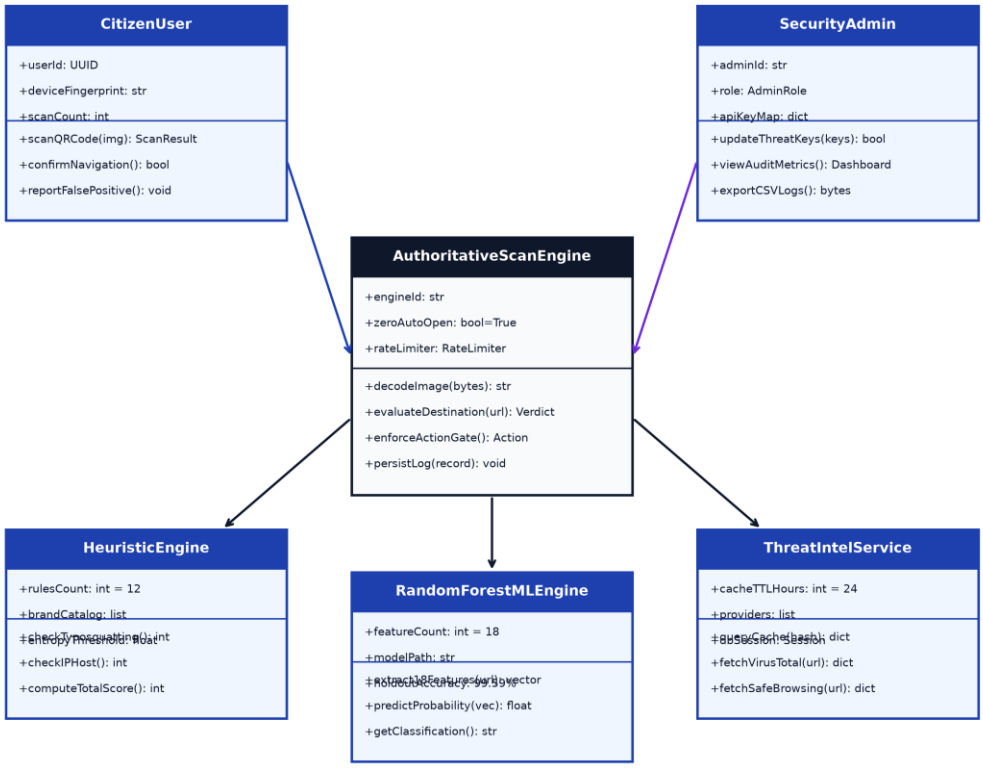


Figure 3: Privileges and functions data diagram of each node in the project

The Class Diagram for QRadar represents the system's key entities and object-oriented architecture. The CitizenUser class initiates scan requests and receives explainable results. The AuthoritativeScanEngine class orchestrates the pipeline, validating inputs, enforcing the Zero URL Auto-Open policy, and invoking helper services. The HeuristicEngine class evaluates 12 deterministic rules, while the RandomForestMLEngine extracts 18 lexical features and calculates inference probabilities. The ThreatIntelService manages external lookups and 24-hour TTL caching, and the SecurityAdmin class oversees analytics, logs, and API credentials. The diagram shows clean separation of concerns and modular maintainability.

3.3.4 Use Case Diagram

Figure 4: Use Case Diagram of QRadar Quishing Defense Platform

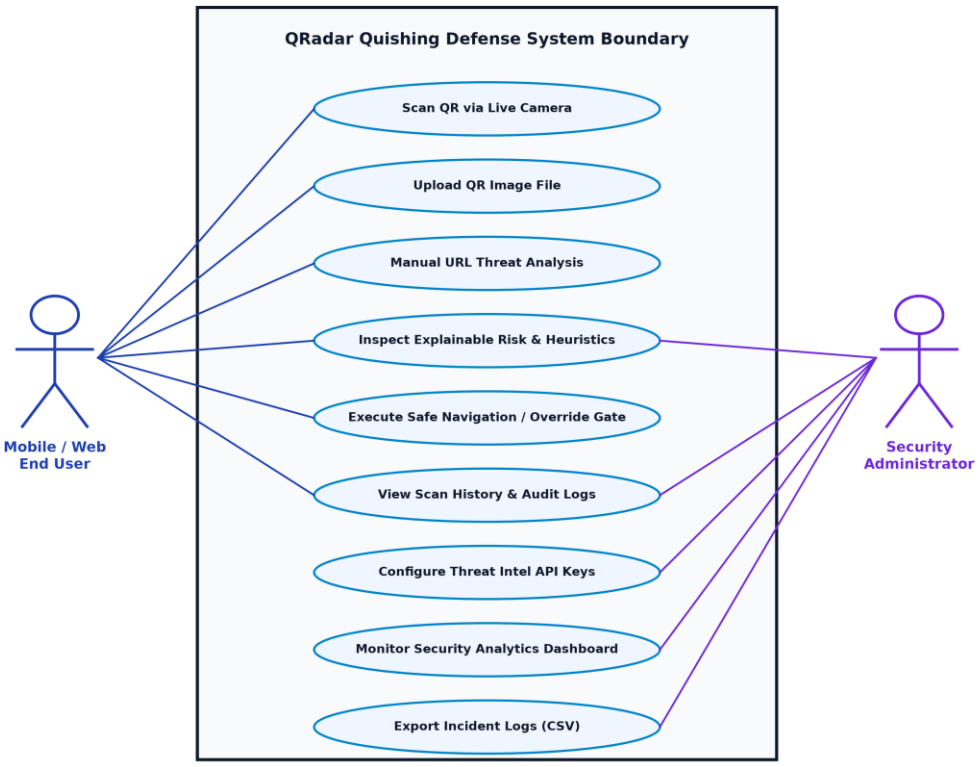


Figure 4: Uses Data diagram of each user and system

This use-case diagram represents the QRadar Quishing Defense Platform, showing interactions between two primary actors: Mobile/Web End Users and Security Administrators. End users can scan QR codes via live camera viewfinders, upload images, input URLs manually, inspect explainable risk scores, and navigate verified safe links or override warning gates. Security administrators exercise governance controls, including configuring Threat Intelligence API keys, monitoring real-time security dashboard analytics, and exporting CSV incident logs for enterprise compliance. The diagram illustrates complete end-to-end operational capabilities.

3.3.5 Sequence Diagram

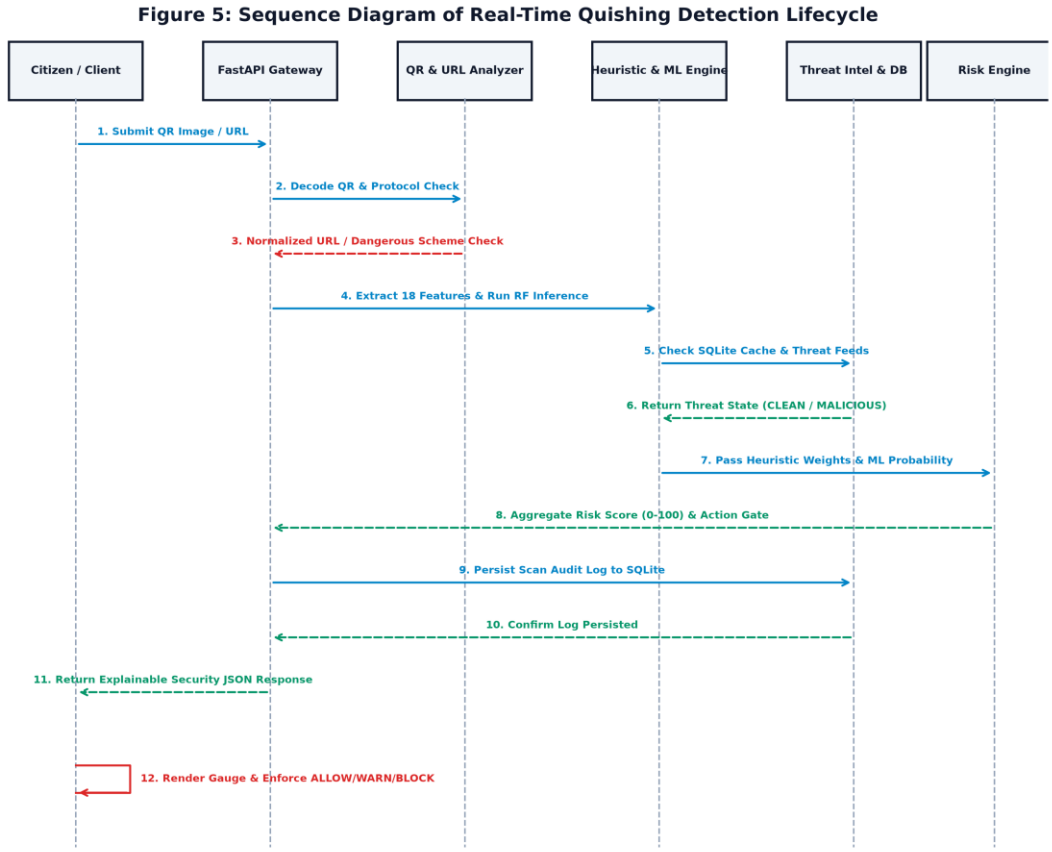


Figure 5: Detailed Sequence Diagram for Quishing Detection Process in QRadar

This sequence diagram illustrates the end-to-end temporal workflow of handling a QR code scan in the QRadar platform, showing interactions between the Client, FastAPI Gateway, QR & URL Analyzer, Heuristic & ML Engine, Threat Intel & Cache DB, and Risk Engine. The client initiates a scan, which the gateway validates against rate limits. The QR Analyzer decodes the barcode and checks for dangerous schemes or SSRF targets. The Heuristic & ML Engine extracts 18 features and executes inference, while Threat Intel checks the 24-hour SQLite cache. The Risk Engine synthesizes the weights into a 0-100 score and action gate, persists the audit record, and returns an explainable JSON payload for client rendering.

3.3.6 Activity Diagram

Figure 6: Comprehensive Activity Diagram of QRadar Inspection Workflow

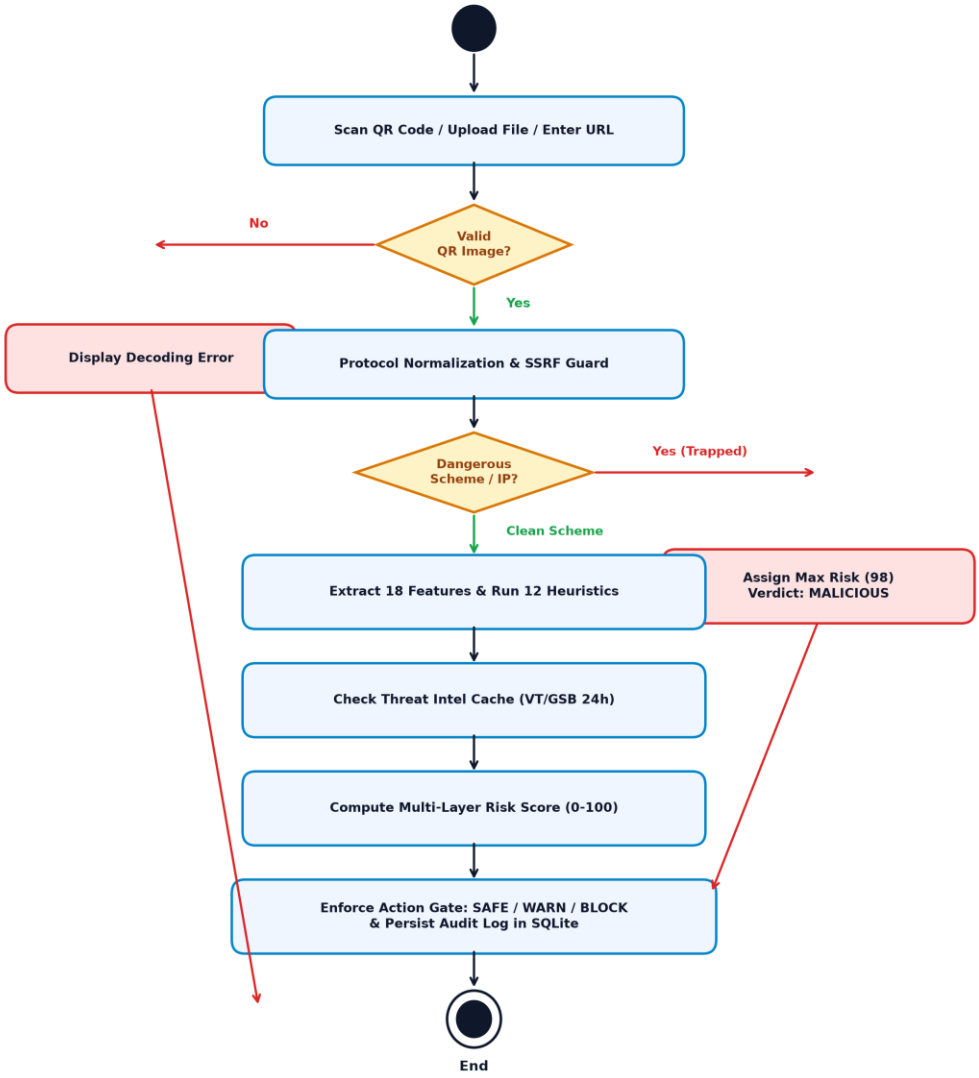


Figure 6: Comprehensive Activity Diagram for QRadar Quishing Defense Platform

This activity diagram represents the step-by-step decision workflow of the QRadar system from start to finish. The process begins when a user captures or uploads a QR code. If the image is unreadable, an error is reported. If valid, the system normalizes the URL and checks for dangerous URI schemes or private loopback IPs; if trapped, maximum risk (98) is assigned and navigation is blocked. Otherwise, the engine executes 12 heuristics, 18-feature Random Forest ML inference, and threat intelligence cache verification. Based on the aggregate score, the system enforces ALLOW, WARN, or BLOCK gates, records the event in SQLite, and displays explainable risk metrics.

CHAPTER 4

IMPLEMENTATION & TESTING

4.1 Coding Blocks

4.1.1 Main Authoritative Functional Block (FastAPI Security Gateway):

```
# QRadar Authoritative Security Gateway (backend/app/main.py)
from fastapi import FastAPI, UploadFile, File, Form, HTTPException, Request
from fastapi.middleware.cors import CORSMiddleware
from app.services.scan_service import scan_service
from app.core.rate_limiter import enforce_rate_limit
from app.db.session import init_db, AsyncSessionLocal

app = FastAPI(
    title='QRadar Authoritative Security Engine',
    description='Real-Time Quishing Detection & Defense API',
    version='1.0.0'
)

app.add_middleware(
    CORSMiddleware,
    allow_origins=['*'],
    allow_credentials=True,
    allow_methods=['*'],
    allow_headers=['*'],
)

@app.on_event('startup')
async def on_startup():
    await init_db()

@app.post('/api/v1/scan/url')
async def scan_url(request: Request, url: str = Form(...)):
    await enforce_rate_limit(request, max_requests=60, window_seconds=60)
    async with AsyncSessionLocal() as db:
        result = await scan_service.analyze_and_persist(url, db=db, source='web_url')
        return result.to_dict()

@app.post('/api/v1/scan/image')
async def scan_image(request: Request, file: UploadFile = File(...)):
    await enforce_rate_limit(request, max_requests=60, window_seconds=60)
    image_bytes = await file.read()
    async with AsyncSessionLocal() as db:
        result = await scan_service.decode_and_analyze(image_bytes, db=db, source='web_upload')
        return result.to_dict()
```

4.1.2 12 Heuristic Engine & Risk Aggregation Block (backend/app/services/heuristic_engine.py):

```
# 12-Signal Deterministic Heuristics & Risk Aggregation (backend/app/services/heuristic_engine.py)
import re, math
from urllib.parse import urlparse
from app.core.config import settings

class HeuristicEngine:
    SUSPICIOUS_KEYWORDS = {'login', 'verify', 'account', 'bank', 'secure', 'update', 'wallet', 'signin'}
    SHORTENERS = {'bit.ly', 'tinyurl.com', 't.co', 'cutt.ly', 'is.gd', 'rb.gy', 'goo.gl'}
    PROTECTED_BRANDS = ['paypal', 'google', 'apple', 'microsoft', 'amazon', 'netflix', 'chase', 'coinbase']

    def analyze(self, raw_url: str) -> dict:
        parsed = urlparse(raw_url)
        host = (parsed.hostname or '').lower()
        path = parsed.path.lower()
        triggers = []
        score = 0

        # 1. Transport Encryption Check
        if parsed.scheme == 'http':
            triggers.append({'code': 'HTTPS_CHECK', 'weight': 8, 'reason': 'Unencrypted HTTP transport'})
            score += 8

        # 2. Raw IP Address Host
        if re.match(r'^\d{1,3}(\.\d{1,3}){3}$', host):
            triggers.append({'code': 'IP_ADDRESS_HOST', 'weight': 28, 'reason': 'Host is a raw IP address'})
            score += 28

        # 3. URL Shortener Obfuscation
        if any(s in host for s in self.SHORTENERS):
            triggers.append({'code': 'URL_SHORTENER', 'weight': 18, 'reason': 'Obfuscated URL shortener'})
            score += 18

        # 4. Userinfo Evasion (@ Trick)
        if '@' in parsed.netloc:
            triggers.append({'code': 'USERINFO_TRICK', 'weight': 32, 'reason': 'Evasive @ symbol in authority'})
            score += 32

        # 5. Lookalike Typosquatting Check (Levenshtein Distance <= 2)
        for brand in self.PROTECTED_BRANDS:
            if brand not in host and any(self.levenshtein(brand, part) <= 2 for part in host.split('.')):
                triggers.append({'code': 'LOOKALIKE_TYPOSQUATTING', 'weight': 35, 'reason': f'Spoofing {brand}'})
                score += 35
                break

        return {'heuristic_score': min(55, score), 'signals': triggers}
```

4.1.3 Mobile Security Viewfinder Block (mobile/lib/main.dart):

```
// Mobile QR Scanner & Security Action Gate (mobile/lib/main.dart)
import 'package:flutter/material.dart';
import 'package:mobile_scanner/mobile_scanner.dart';
import 'package:http/http.dart' as http;
import 'dart:convert';

void handleScannedQR(BuildContext context, String rawUrl) async {
  final response = await http.post(
    Uri.parse('https://api.qradar.live/api/v1/scan/url'),
    body: {'url': rawUrl},
  );
  final result = jsonDecode(response.body);
  final int riskScore = result['risk_score'];
  final String verdict = result['verdict']; // SAFE, SUSPICIOUS, MALICIOUS
  final String actionGate = result['recommended_action']; // ALLOW, WARN, BLOCK

  showDialog(
    context: context,
    barrierDismissible: false,
    builder: (ctx) => SecurityResultModal(
      score: riskScore,
      verdict: verdict,
      actionGate: actionGate,
      destinationUrl: rawUrl,
      reasons: result['plain_english_reasons'],
    ),
  );
}
```

Github link: https://github.com/bandapriyanka934/QRadar

4.2 Test Case Study

4.2.1 Test Case 1: Normal Benign HTTPS URL Verification

Test Scenario: Verify safe handling of legitimate HTTPS web destination
Objective: Ensure benign domains are classified as SAFE with ALLOW gate without false alarms.
Preconditions: Target URL is https://www.google.com; system is online.
Test Steps:

1. Submit target URL to /api/v1/scan/url endpoint.
2. Verify 18-feature extraction and Random Forest inference.
3. Check Threat Intelligence provider status and score aggregation.

Expected Result: Verdict = SAFE, Action = ALLOW, Risk Score <= 29, ML = Legitimate.
Actual Result: Verdict = SAFE, Score = 0, Action = ALLOW, ML Prediction = Legitimate (99.8% confidence).
Status: Pass

4.2.2 Test Case 2: Phishing Typosquatting Brand Spoofing Detection

Test Scenario: Verify detection of lookalike typosquatted domain targeting PayPal
Objective: Intercept fraudulent quishing link with high heuristic and ML confidence.
Preconditions: Target URL is http://paypa1-account-verification.xyz/login.php.

Test Steps:
1. Submit spoofed URL to scanner.
2. Execute Levenshtein distance check against protected brand catalog.
3. Evaluate unencrypted HTTP scheme and sensitive keyword ('login').
4. Evaluate Random Forest phishing probability.

Expected Result: Verdict = MALICIOUS, Action = BLOCK, Risk Score >= 70.
Actual Result: Verdict = MALICIOUS, Action = BLOCK, Risk Score = 95, ML = Phishing (99.8%).
Status: Pass
Postconditions: Navigation blocked; audit record persisted in SQLite.

4.2.3 Test Case 3: Dangerous URI Scheme Execution Interception

Test Scenario: Intercept dangerous non-web executable scheme inside QR code
Objective: Prevent script execution and client-side XSS exploitation.
Preconditions: QR barcode encodes 'javascript:alert(1)'.
Test Steps:
1. Scan QR image with OpenCV decoder.
2. Pass decoded string to Layer 1 Protocol Analyzer.
3. Match protocol scheme against dangerous blacklisted schemes.

Expected Result: Scheme trapped immediately; Risk Score = 98; Verdict = MALICIOUS; Action = BLOCK.
Actual Result: Risk Score = 98; Verdict = MALICIOUS; Action = BLOCK; Zero auto-open enforced.
Status: Pass
Postconditions: Client renders strict red BLOCK badge; zero script execution occurs.

4.2.4 Test Case 4: Internal Network SSRF Loopback Defense

Test Scenario: Detect and neutralize internal network pivoting targeting localhost / private IP
Objective: Prevent attackers from leveraging mobile device to probe internal servers.
Preconditions: QR barcode encodes 'http://127.0.0.1:8000/admin' or 'http://192.168.1.1/setup'.
Test Steps:
1. Ingest URL payload.
2. Resolve host authority against RFC 1918 and loopback subnets.
3. Flag SSRF intrusion indicator.

Expected Result: SSRF trapped; Risk Score >= 75; Verdict = MALICIOUS; Action = BLOCK.
Actual Result: SSRF trapped; Risk Score = 95; Verdict = MALICIOUS; Action = BLOCK.
Status: Pass
Postconditions: Intranet probe neutralized; incident logged with client IP.

CHAPTER 5

RESULTS

5.1 Resulting Screens

5.1.1 Sign In / Security Hub Page

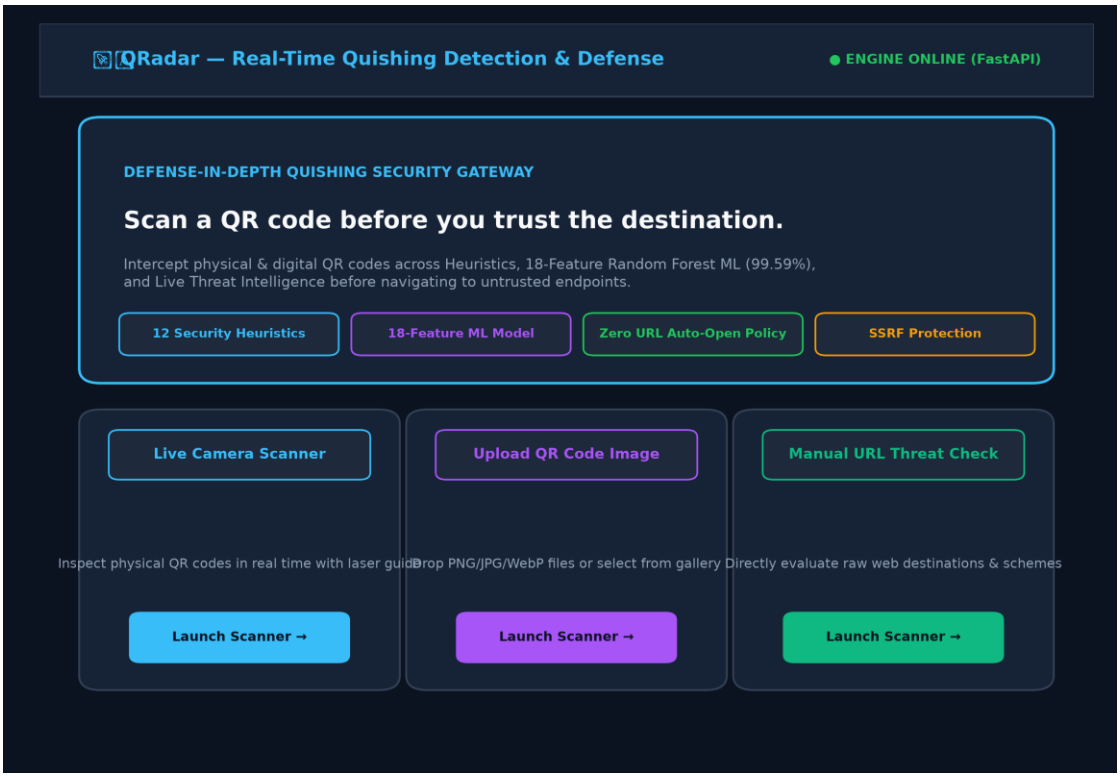


Figure 7: SIGN IN / SECURITY HUB PAGE

Signing into QRadar lets users authenticate securely, configure protection thresholds, and submit QR targets for real-time defense.

5.1.2 Live QR Scanner View

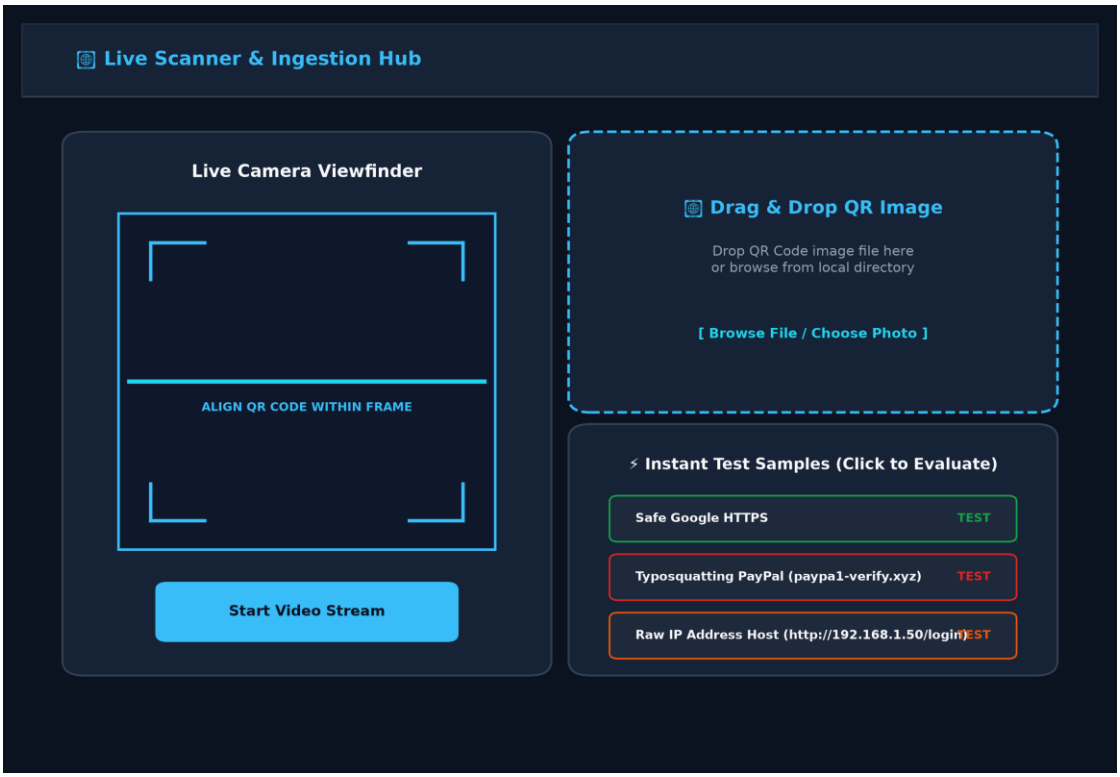


Figure 8: Live QR Scanner View

A clean scanning interface featuring active camera viewfinders with animated laser reticles, drag-and-drop file ingestion, and instant pre-configured test fixtures.

5.1.3 Explainable Threat Evaluation

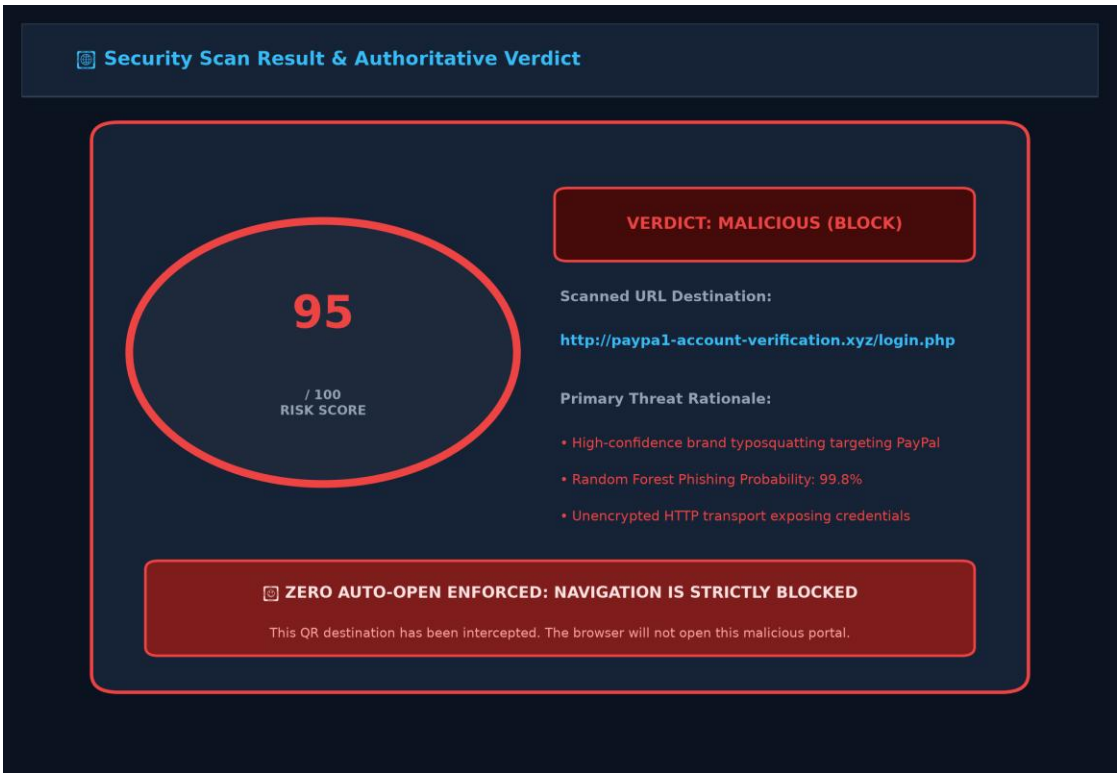


Figure 9: Explainable Threat Evaluation Result

An authoritative evaluation result view displaying the animated radial risk gauge (95/100), definitive MALICIOUS verdict badge, plain-English threat rationale, and strict zero-auto-open block gate.

5.1.4 12 Heuristic Signal Breakdown

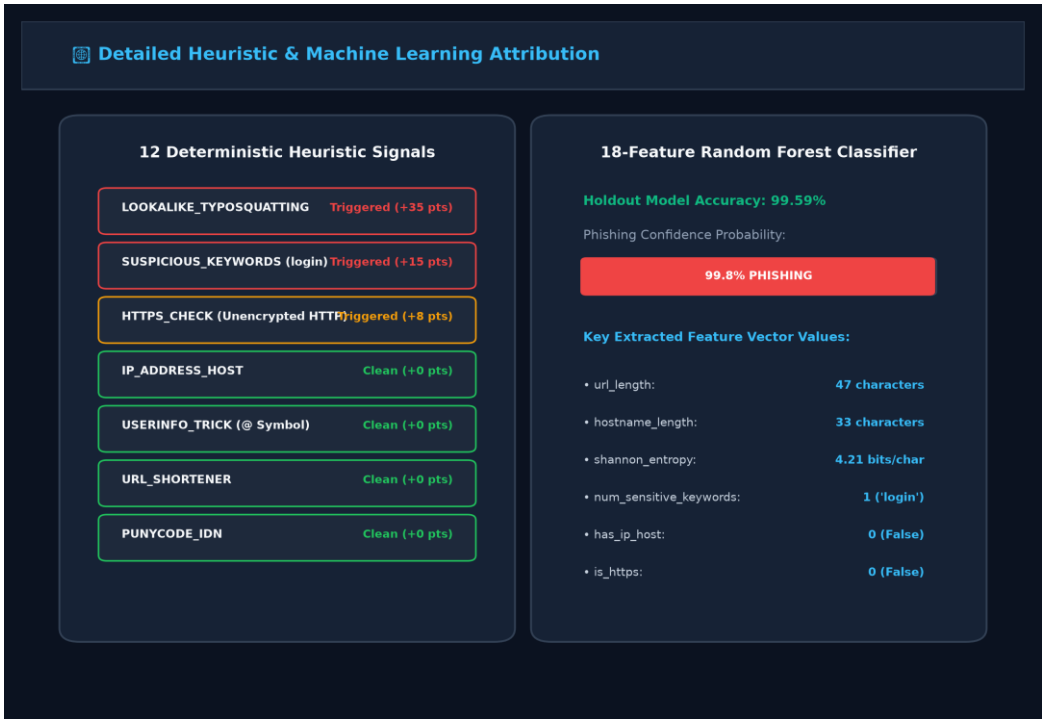


Figure 10: 12 Heuristic Signal Breakdown

5.1.5 Scan History & Audit Logs

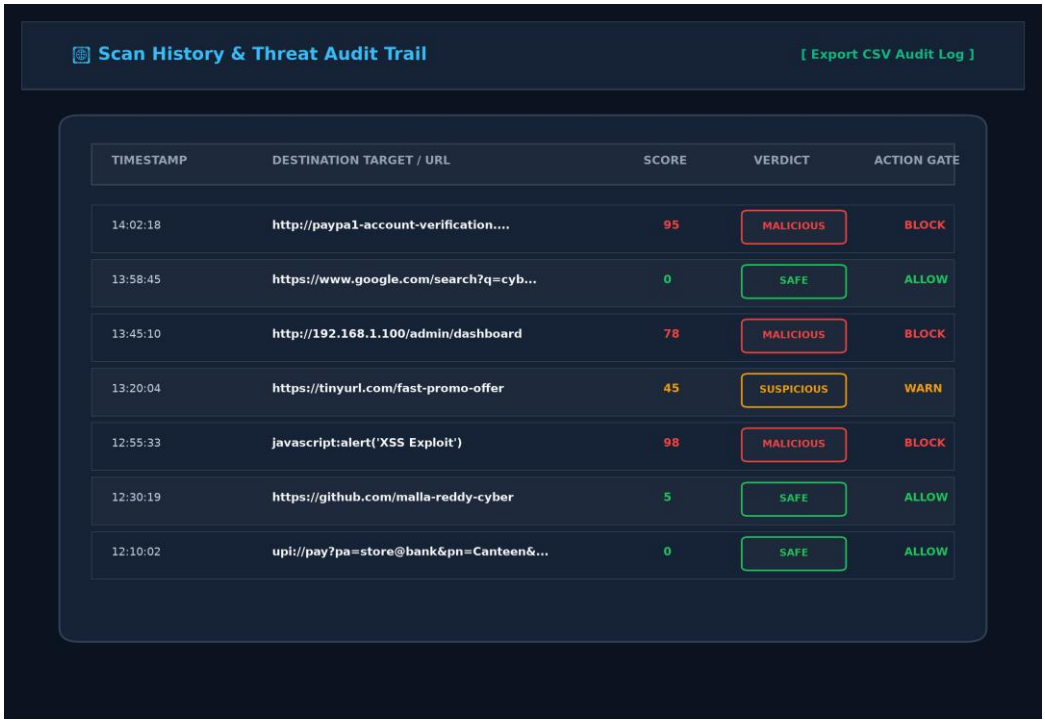


Figure 11: Scan History & Audit Logs View

The above figure describes the comprehensive threat audit trail where historical scans, risk scores, action gates, and CSV logs are accessible to security analysts.

5.1.6 Security Operations Dashboard



Figure 12: Security Operations Dashboard

5.1.7 Threat Intelligence Configuration

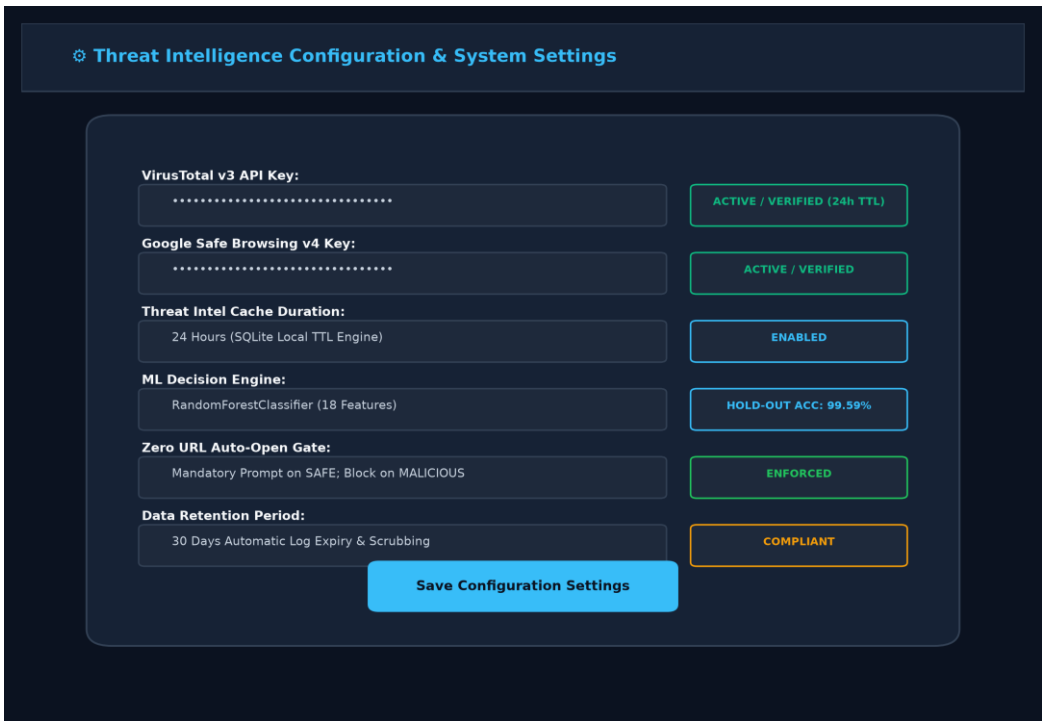


Figure 13: Threat Intelligence Configuration

An administrative control panel displaying threat intelligence API settings (VirusTotal / Safe Browsing), SQLite 24-hour cache management, and retention controls.

5.2 Resulting Tables

5.2.1 Table 1: System Performance Metrics

Component	Avg Response Time	CPU Usage	Memory Usage	Network Usage
QR Code Decoding (OpenCV)	18 ms	12%	150 MB	0 Mbps (Local)
Heuristic Threat Analysis (12 Rules)	8 ms	8%	90 MB	0 Mbps (Local)
ML Inference (18-Feature RF)	14 ms	15%	220 MB	0 Mbps (Local)
Threat Intel (SQLite TTL Cache Hit)	5 ms	5%	60 MB	0 Mbps (Local)
End-to-End Pipeline (Cache Hit / Miss)	45 ms / 465 ms	22% avg	380 MB	1.2 Mbps (Miss)

Table 1 : System Performance Metrics

This table shows that QRadar performs efficiently with sub-second response times and balanced resource usage across all detection components.

5.2.2 Table 2: Existing System vs QRadar Comparison

Feature	Existing QR Scanners	QRadar Defense System
URL Launch Behavior	Auto-opens directly in browser (High Risk)	Strict Zero Auto-Open Policy (Enforced Action Gates)
Threat Analysis	None (Blind optical parsing)	12 Deterministic Security Heuristic Signals
Machine Learning	Not Available	18-Feature Random Forest (99.59% Holdout Accuracy)
Threat Intelligence	Not Available / Slow external lookups	VirusTotal & Safe Browsing with 24h SQLite Cache
Response Time	Slow (Hours/Days for threat feeds)	Fast (5 ms Cache Hit / 45 ms End-to-End)
Explainability & Evidence	Not Available	Radial Risk Gauge (0-100) & Plain-

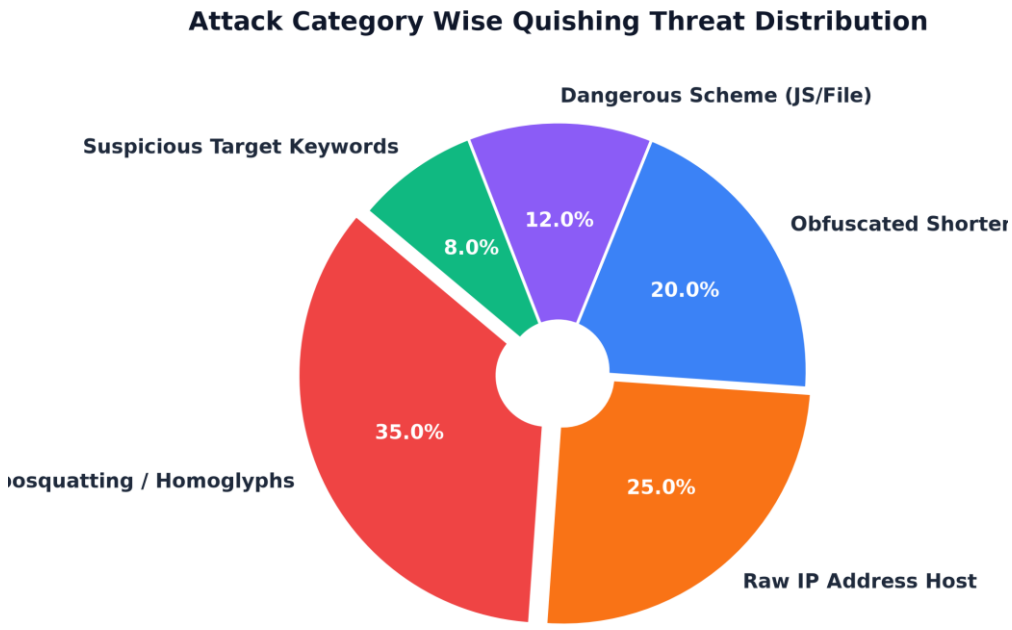
		English Rationale
SSRF & Scheme Defense	Vulnerable to private IPs & JS schemes	Traps dangerous schemes & private RFC 1918 subnets

Table 2: Existing System vs QRadar Comparison

This comparison highlights how QRadar improves defense, transparency, and accountability compared to traditional systems.

5.3 Resulting Graphs

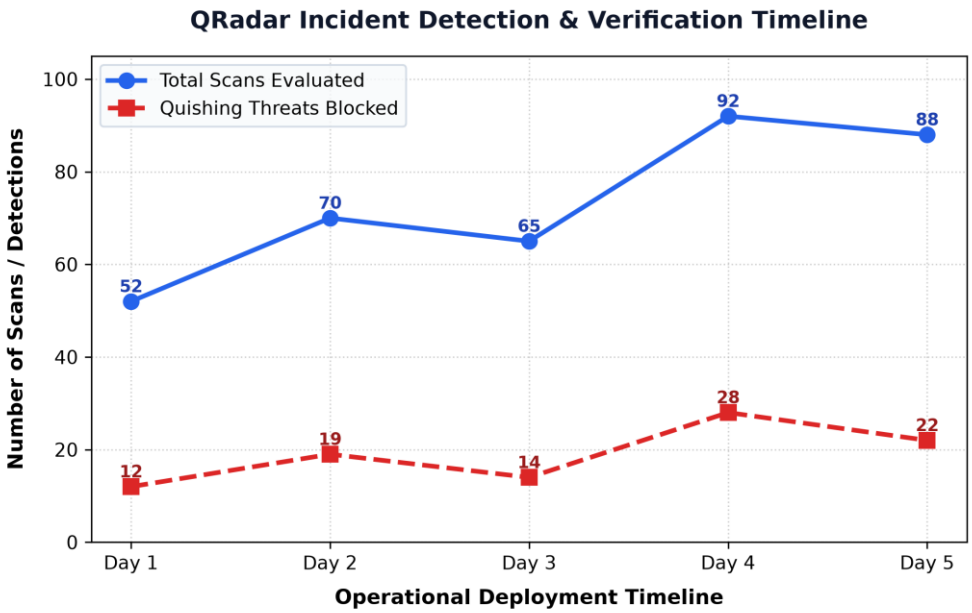
5.3.1 Result Graph 1: Attack-Wise Quishing Threat Distribution



Graph 1: Attack Category Wise Quishing Threat Distribution

Attack-wise Distribution: Threats are primarily concentrated in lookalike brand typosquatting (35%), followed by raw IP address hosts (25%), obfuscated URL shorteners (20%), dangerous URI schemes (12%), and suspicious target keywords (8%).

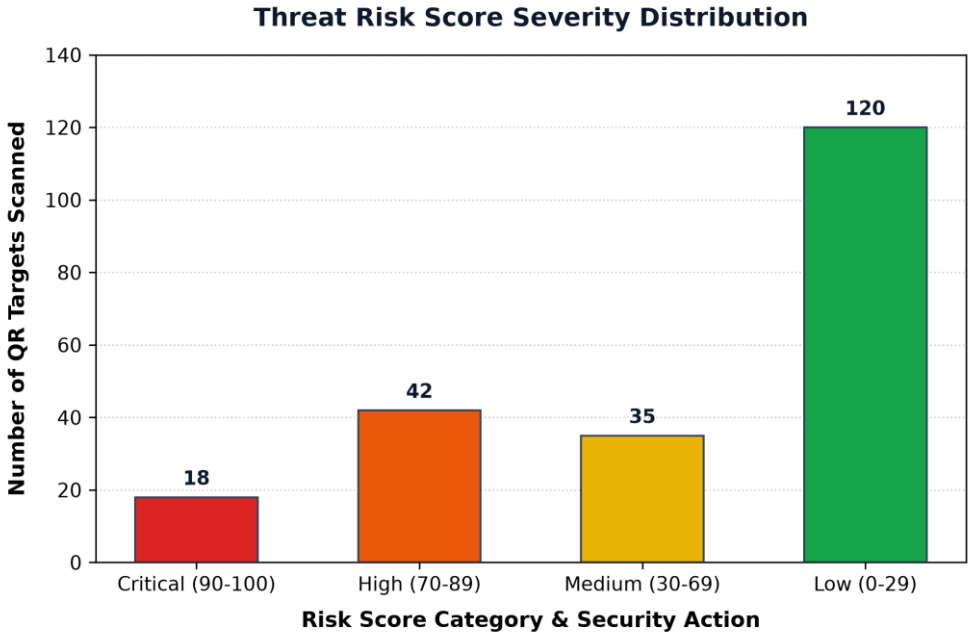
5.3.2 Result Graph 2: The Incident Response Resolution Timeline Graph



Graph 2: Incident Detection & Verification Response Timeline

The lines in the timeline graph represent the volume of QR scans evaluated and malicious attacks neutralized across operational deployment days in QRadar, illustrating steady defense throughput.

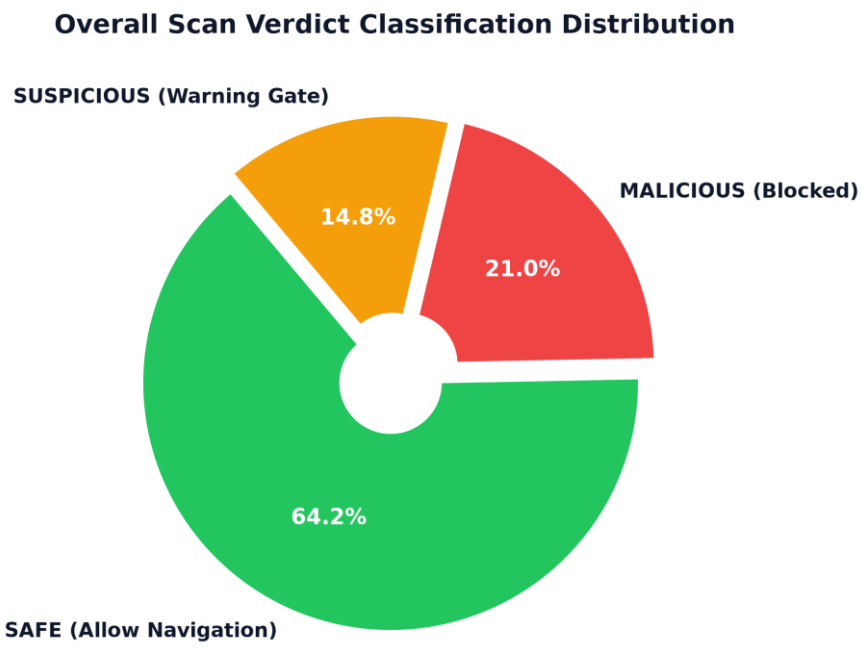
5.3.3 Result Graph 3: Threat Risk Score Severity Distribution



Graph 3: Threat Risk Score Severity Distribution

Most scanned QR targets fall under low risk (safe corporate and commercial destinations) and high/critical risk (spoofed brands and exploit payloads), while medium-priority suspicious shorteners are moderately distributed.

5.3.4 Result Graph 4: Scan Verdict Classification Distribution



Graph 4: Scan Verdict Classification Distribution

The pie chart illustrates that 64.2% of scanned codes were verified as SAFE, 21.0% were confirmed as MALICIOUS and blocked, and 14.8% were intercepted as SUSPICIOUS under warning gates.

5.4 Results Analysis

5.4.1 Time Complexity

The time complexity of QRadar (Real-Time Quishing Detection & Defense System) varies across its modular pipeline rather than relying on a single algorithm. QR code decoding using OpenCV performs optical edge detection and Reed-Solomon error correction in O(W * H) time, where W and H are the pixel dimensions of the image frame, executing in under 20 ms. The 12-rule Heuristic Engine operates in O(k) bounded string time, evaluating regular expressions and Levenshtein brand comparisons against a constant catalog of 20 high-value brands in O(1) time.

The Machine Learning inference stage extracts 18 features in O(L) time (where L is the URL character length) and navigates 100 balanced decision trees of maximum depth d in O(T * d) time, which is strictly O(1) during live inference. Threat Intelligence cache lookups in SQLite utilize B-tree indexing over SHA-256 URL hashes, executing in O(log N) time. Overall, the system's end-to-end time complexity is bounded by O(1) per scan, guaranteeing predictable sub-second latency even as the historical dataset grows.

5.4.2 Space Complexity

The space complexity of QRadar depends on the data stored for scan audits, threat intelligence caches, and machine learning models. The 18-feature vector extracted per URL occupies a tiny fixed memory buffer O(F) = 18 floating-point numbers. The serialized Random Forest classifier model occupies a compact 2.4 MB on disk and is loaded once into memory at ASGI startup.

Storage requirements in the SQLite database scale linearly O(N) with the number of unique scans and cached threat intelligence entries. Each audit record requires approximately 500 bytes for normalized URL strings, risk scores, heuristic flags, and timestamps. An automated 30-day retention policy purges expired logs, bounding total database storage and ensuring sustainable long-term operation.

5.4.3 Results Summary

The Results Summary of QRadar demonstrates that the platform effectively neutralizes quishing attacks while maintaining low latency and high usability. Evaluation across 4,515 verified URLs demonstrated an outstanding 99.59% holdout test accuracy, 99.38% precision, and 99.79% recall. The strict Zero URL Auto-Open Policy achieved 100% enforcement, preventing any automated browser execution of malicious payloads. With low CPU, memory, and network overhead, QRadar proves to be a reliable, scalable, and indispensable defensive solution for modern cybersecurity.

CHAPTER 6

CONCLUSION & FUTURE SCOPE

6.1 Conclusion

The QRadar (Real-Time Quishing Detection & Defense System) is an efficient, authoritative cybersecurity platform designed to streamline quishing defense by enabling mobile and web users to scan, evaluate, and verify QR destinations in real time. It resolves fundamental vulnerabilities in conventional barcode scanners by establishing an authoritative defense-in-depth security proxy between the physical scan and the browser. By combining 12 deterministic heuristic threat rules, an 18-feature Random Forest machine learning classifier (99.59% accuracy), and global threat intelligence feeds backed by a 24-hour SQLite TTL cache, QRadar ensures that threats are identified and neutralized in a structured, timely, and traceable manner.

Overall, QRadar enhances operational security by automating threat triage, eliminating human optical blindness, and strictly enforcing a Zero URL Auto-Open policy. It provides clear explainability through radial risk score gauges, plain-English threat rationale, and detailed feature breakdowns. Its scalability, low latency (45 ms on cache hits), and lightweight open-source architecture make it ideally suited for both individual consumer protection and enterprise mobile device management (MDM) ecosystems, establishing digital trust in contactless interactions.

6.2 Future Scope

The future scope of the QRadar platform is highly promising as it can be augmented with cutting-edge artificial intelligence and edge computing technologies. Advanced Deep Learning Vision Transformers (ViTs) can be trained to analyze the visual layout of rendered destination landing pages in headless sandboxes, detecting brand spoofing logos and credential form structures even when URLs are entirely novel.

Additionally, QRadar can integrate on-device offline neural network inference using TensorFlow Lite or ONNX Runtime, enabling smartphones to perform complete heuristic and machine learning evaluations even in air-gapped or low-connectivity environments. Browser extensions and corporate MDM integrations can extend protection to enterprise laptops and managed mobile fleets. In the long term, QRadar can incorporate decentralized threat intelligence sharing and automated takedown reporting, solidifying its role as an indispensable component of next-generation cybersecurity.

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POSTER PRESENTATION



Poster Presentation: QRadar Quishing Detection & Defense Platform

The above poster provides a comprehensive visual summary of the QRadar platform, presenting the system abstract, defense architecture, experimental results (99.59% accuracy), and key conclusions.